Lesson 8.5 Guided Notes

Session 1 · Graphing Inequalities

I can graph an inequality on a number line using an open or a closed dot. · 6.EE.B.8

Word / Spanish	Meaning — complete the blank	Example
boundary point / punto límite	The number where the solutions _____ or stop.	the 8 in x ≥ 8
closed circle / círculo cerrado	A filled dot: the boundary _____ a solution.	used for ≤ and ≥
open circle / círculo abierto	An empty dot: the boundary is _____ a solution.	used for < and >
solution set / conjunto solución	_____ the numbers that make an inequality true.	x ≥ 8: 8, 9.5, 20, …

Key idea

Use a _____ dot for ≤ and ≥. Use an _____ dot for < and >.

Draw the arrow toward the numbers that make the inequality _____.

Worked example

Graph x < 6 on a number line.

The boundary number is _____.

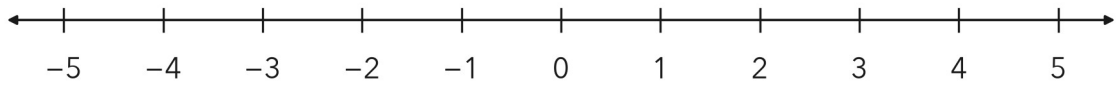
< does not include 6, so draw an _____ dot at 6.

x is less than 6, so the arrow points _____.

Answer: Check: 5 < 6 is _____, and 5 is under the arrow.

You try

1. Graph $y \geq -2$ on the number line.



_____

2. Write the inequality for each graph, using x.
a. closed dot at 4, arrow left _____
b. open dot at −1, arrow right _____

_____

Sum it up: I choose an open or a closed dot by looking at ___, and I point the arrow ___.

_____

Session 2 · Testing Solutions to Inequalities

I can test values against an inequality and graph the solution set. · 6.EE.B.8

Word / Spanish	Meaning — complete the blank	Example
substitute / sustituir	Put a number in _____ of the variable.	x = 9 in x > 5 gives 9 > 5
solution / solución	A value that makes the inequality _____.	9 is a solution of x > 5
true statement / enunciado verdadero	A comparison that is _____.	9 > 5

Key idea

To test a value, substitute it and ask whether the statement is _____.

For ≤ and ≥, the boundary _____ a solution. For < and >, it is _____.

Worked example

Test 8, 9 and 10 in x < 9.

8 < 9 is _____, so 8 is a solution.

9 < 9 is _____, so the dot at 9 is _____.

10 < 9 is false, so 10 is not a solution.

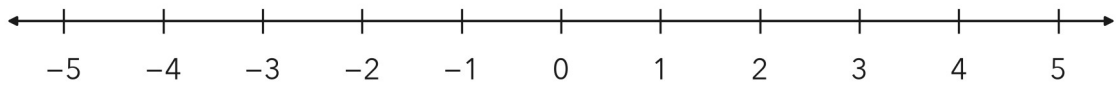
Answer: Only _____ works. Graph: open dot at 9, arrow left.

You try

1. Is the value a solution? Write yes or no.
a. $y \geq 3$; $y = 3$ _____
b. $m < -2$; $m = 0$ _____

_____

2. Test −1, 2 and 5 in $x > 2$. Then graph $x > 2$.



_____

Sum it up: To decide whether a number is a solution, I ___, and the boundary is a solution only when ___.

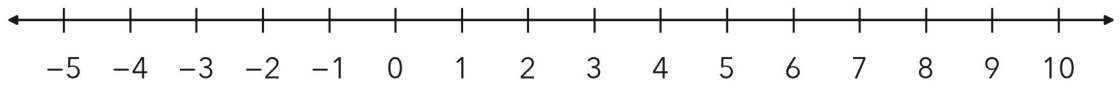
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Session 1 · Model, reason, and explain

Graphing Inequalities · 6.EE.B.8

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Graph x≥−2. Test −3,−2,0 to choose the endpoint and direction. Explain what a closed circle tells the reader.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: chooses arrow direction from the symbol’s appearance without checking a value.. Explain what changes and what stays true.

_____

_____

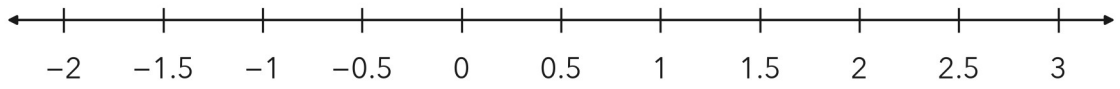
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Testing Solutions to Inequalities · 6.EE.B.8

Success: represent the quantities, justify the relationship, and check the result with the correct units.

For x<1.5, test −1,1.5,2. Give another fractional solution and explain why the set is not limited to tested values.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: reports only the listed candidates as the complete solution set.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?